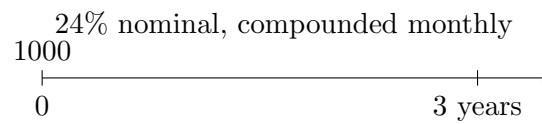


Problem 9.1

A man borrows \$1,000 at an interest rate of 24% per year compounded monthly. How much does he owe after 3 years?

Solution

The loan can be represented on a timeline as follows:



A nominal annual interest rate of 24% compounded monthly has monthly rate

$$\frac{0.24}{12} = 0.02.$$

In 3 years, there are

$$12(3) = 36$$

monthly compounding periods. Therefore, the accumulated value is

$$1000 \left(1 + \frac{0.24}{12} \right)^{12 \cdot 3} = 1000(1.02)^{36} \\ \approx 2039.887344.$$

Thus, after 3 years he owes

$$\boxed{\$2,039.89}.$$